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Project Title:	DNA Damage Clusters in Human Cell Transformation Induced by Single or Multiple Space Radiation Ion Exposures		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Zip Code:	11973-5000	**Congressional District:**	1
Comments:	PI Deceased as of October 2009.		
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
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Flight Program:			
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Task Description:

The space radiation environment poses long and short-term risks to human travelers. During a 3-year Mars mission, the nucleus of each cell in an astronaut's body will be hit by ~400 protons and 0.4 high energy, high atomic number (HZE) ions, plus high levels of protons during Solar Particle Events (SPEs). Thus, understanding the biological consequences of multiple particle exposures is essential for assessing radiation risks of space travel. Our research focuses on understanding the consequences to human cells of exposure to the levels and kinds of space radiations present in deep space. We use a human primary cell neoplastic transformation model, in which irradiated human cells acquire growth properties of human cancer cells. We found that exposure to a moderate proton dose followed by a HZE particle gave higher transformation levels than the sum of the two exposures, indicating non-linear effects of multiple radiation exposures. We now propose to expose human cells to multiple space ions including protons at low doses or in SPE-simulating proton fields and HZE ions and measure cell survival and transformation.
Space radiation induces complex DNA damages affecting both DNA strands/double strand breaks (DSBs) and clusters of damaged bases. We plan to measure the induction, repair and levels of unrepaired clusters and DSBs, and to correlate these data to killing and transformation.

Countermeasures to reduce DNA and cellular damage are important in reducing space travel risk. Melatonin, a human hormone, effectively scavenges radiation-induced hydroxyl radicals that damage DNA. Our preliminary data indicate that melatonin effectively reduces cell transformation by HZE ions. We propose to test its effectiveness in reducing biological damage induced in human cells by single or multiple space particles.

Rationale for HRP Directed Research:

Research Impact: We are addressing Research & Technology Question 28a, (the effect of radiation quality and dose rate on carcinogenesis probability), using four hypotheses directly addressed by our four specific aims. These are: Hypothesis 1: Low proton fluencies will not induce cell killing, but may affect cell cycle progression and thus could affect transformation by later HZE particle hits. Specific Aim 1: Determine the effects, including transformation, induced by realistic space radiation exposures of protons (1, 2, or 4 protons per cell nucleus).

Hypothesis 2: protons of the energy and LET (linear energy transfer) ranges present in SPEs will have similar effects on cell survival, transformation and DNA damage spectrum as high energy (1 GeV/n) protons. Specific Aim 2: Measure transformation, cell killing and DNA damage spectra and repairability of complex damages produced by protons of different energies in SPEs.

Hypothesis 3: SPE beams will affect cell killing and transformation as the sum of the constituent photons at the same dose rate. Specific Aim 3: Compare biological effects of SPE beams with the sums of protons of similar energies and dose rates (and to 1 GeV/n protons at acute dose rates).

Hypothesis 4: Mixed field irradiations of 1, 2 or 4 H/nucleus followed by one HZE hit/nucleus will produce survival curves dominated by HZE effect; however, low H fluxes may induce repair and thus decrease transformation frequencies. Specific Aim 4: Determine the transforming and killing potential of dual hits by 1, 2 or 4 protons/nucleus subsequently followed by one HZE particle/nucleus.

Research Impact/Earth Benefits:

Earth Benefits: This project is determining the effects of charged particles on human cells, specifically the killing properties of the radiation and its ability to induce cell transformation, a model for human cancer. Data obtained in this grant year show that both the killing ability and the transforming ability of the radiation increase with the amount of energy deposited by the specific type of particle, known as the radiation's linear energy transfer characteristic. These studies are directly applicable to life on earth, as charged particles are beginning to be used in tumor radiotherapy in humans. The advantage of particles over gamma rays or X-rays is better targeting of tumors and less damage to adjacent non-tumor tissue. Protons for cancer therapy are now available at several US medical centers and additional proton radiotherapy facilities are under construction or being planned. In addition, heavier particles such as carbon ions are in use in Europe and in Japan for radiation therapy, and offer the possibility of even better spatial targeting of tumors than is possible with protons. Concern has arisen concerning additional radiation generated by activation of material in the treatment room by the particles during their use for radiation therapy. Similar activation occurs in the target room during irradiation of experimental samples; this offers an excellent opportunity to characterize this adventitious radiation and to devise approaches to minimize it.

Innovative Technology For investigating differences in repair of X-ray vs. charged particle-induced DNA damages, we designed and developed a new dual-fluorescence tag system for visualizing and quantifying DNA repair. This assay replaces the use of radioactive DNA labeling with fluorescent labels, thus greatly decreasing preparation frequency and thus experimental costs as well as avoiding the use of radioactivity and eliminating disposal costs. This system won a Department of Energy Best in Class Pollution Prevention Award. Results from the use of this new technology to probe repair of space radiation-induced DNA damage are now in press in Radiation Research.

During the first year of this project, we are investigating five aspects of human cell transformation by space radiation and the ability of cells to repair space radiation damage: 1) effects of low proton fluences on human cell neoplastic transformation, 2) mechanisms of increased transformation by proton plus heavy charged particle irradiation, 3) efficiencies of different charged particles in inducing cell transformation, 4) effectiveness of simulated Solar Particle Event radiation in killing or transforming human cells, and 5) induction and repair of complex DNA damages in human cells.
In all these experiments, we use primary human skin cells as the test biological sample. Cells are exposed to radiation at the NASA Space Radiation Laboratory at Brookhaven National Laboratory. We then determine the fraction of cells surviving the radiation exposure that can grow and form colonies on plastic plates; we also determine the number of cells that can grow in semi-solid medium, in which normal cells do not grow but cancer cells do grow, forming "anchorage-independent colonies." We count the colonies on the plastic plates and in the semi-solid medium, and calculate the number of transformed colonies per surviving cell. We measure induction of DNA damages by space radiation in human cells, as well as the cells' ability to repair such damage. We also use a model DNA system that allows the examination of specific DNA damages by human DNA repair enzymes.

Effects of Low Proton Fluences on Human Cell Neoplastic Transformation

Task Progress:	In routine space conditions, cells are “hit” only rarely by a proton. Thus most proton exposures to a cell in a human traveler consist of a single proton hit. Our initial experiments were confounded by activation of materials in the target room by previous proton experiments in the same beam session. This resulted in adventitious radiation whose effect overwhelmed that of the intended radiation. We found that by carrying out the low proton dose experiments the first thing in the morning (with the target room unused since the previous day), apparently linear biological responses as a function of proton dose were produced. Further experiments are in progress. Mechanisms of Increased Transformation by Proton Plus Heavy Charged Particle Irradiation We found previously that the timing between exposure of human cells to protons and to a single heavy charged particle (dual hit or mixed-field irradiation) was critical—if this time window was less than about an hour, the transformation efficiency was increased by about 3-fold. We had eliminated two plausible mechanisms: altering the cell cycle and thus radiation sensitivity, and changing the survival of the cells. We are now investigating a new hypothesis—that the initial proton exposure induces many difficult to repair DNA damages, and the cellular repair proteins bind to these damages and are sequestered in these repair-resistant “foci.” Thus fewer repair enzymes would be available to deal with the DNA damages induced by the subsequent heavy charged particle. Efficiencies of Different Charged Particles in Inducing Cell Transformation Understanding the effects of space radiation on human cells requires that we understand the efficiency of different heavy charged particles in inducing cell transformation. This will be especially essential in evaluating the mixed field irradiations that simulate space radiation. We found that the particles that deposit more energy (per particle) are more effective in inducing cell transformation, and the relation between energy deposition and transformation efficiency is approximately linear. We also found that Fe ions (therefore with the same Z or atomic number) that deposit more energy when they traverse the cell also have higher efficiencies of cell transformation. Since protons of different energies deposit rather similar amounts of energy along their path in traversing the cell, it was not clear whether they would have different efficiencies of inducing transformation. However, we found that protons that deposit more energy as they traverse the cell also have increased efficiency of induction of transformation. Effectiveness of Simulated Solar Particle Event Radiation in Killing or Transforming Human Cells We are undertaking initial experiments simulating solar particle event (SPE) exposures to human cells. Since the NASA Space Radiation Laboratory has not previously supplied SPE-simulating radiation for biological experiments, we have planned an initial simple experiment. In this experiment, we plan to use 50 MeV/n protons only, and a dose rate of ~ 1 cGy//min, similar that that found in a typical SPE. We will determine only cell survival in the initial experiments, as realistic SPE doses are compatible with observation of significant loss of cell survival, and because these experiments are technically straightforward. Induction and Repair of Complex DNA Damages in Human Cells. We are investigating the induction and repair of complex DNA damages in human cells and in human tissue. We are irradiating samples with Fe ions, C ions and protons as well as X-ray controls. We irradiate samples with increasing doses of each ion and harvested them immediately to construct a dose-response relation and also irradiate samples at a limited number of doses, then incubate them at 37oC to allow repair. These samples are now being analyzed. We devised a non-radioactive, two fluorescence color system for studying repair of specific clustered damages. This system uses a fluorescent DNA tag rather than a a radioactive one, and won a Department of Energy Best in Class Pollution Prevention Award. We are using this model system to compare the repair of simple vs. complex clustered DNA damages, and plan to compare them with the actual properties of repair of complex damages induced by X-rays vs. those induced by particles.
Bibliography Type:	Description: (Last Updated: 04/12/2011)
Abstracts for Journals and Proceedings	Sutherland BM. "Repair of Clustered DNA Damages Induced by Low Radiation Doses in Human Cells and Skin." Clustered and tandem DNA lesions (symposium), International Congress of Radiation Research, San Francisco, California, July 8-12, 2007. Abstracts, International Congress of Radiation Research, 2007, p. 44. , Jul-2007
Abstracts for Journals and Proceedings	Sutherland BM. "Efficiency of Transformation of Human Primary Cells to Anchorage-Independent Growth by HZE Particles or Protons." Presented at the 18th Annual NASA Space Radiation Health Investigators’ Workshop, Sonoma CA, July 13-15, 2007. Workshop abstracts, 18th Annual NASA Space Radiation Health Investigators’ Workshop, July 2007. , Jul-2007
Abstracts for Journals and Proceedings	Das B, Bennett PV, Sutherland BM. "Bi-Stranded Clustered DNA Damage Induced in Human Cells by Low and High LET Ionizing Radiation." 8th Meeting of the Indian Society of Radiation Biology (ISRB) and the 5th LOWRAD (Low Radiation) International Conference on Low Dose Radiation Effects on Human Health and Environment, Banaras Hindu University, Varanasi, India, November 20-22, 2006. Conference abstracts, 8th Meeting of the Indian Society of Radiation Biology (ISRB) and the 5th LOWRAD (Low Radiation) International Conference on Low Dose Radiation Effects on Human Health and Environment, Banaras Hindu University, Varanasi, India, November 20-22, 2006. , Nov-2006
Abstracts for Journals and Proceedings	Das B, Bennett PV, Kennedy AR, Sutherland BM. "Protective effect of antioxidants against heavy ion induced complex bi-stranded damage clusters in human cells." Conference of the Free Radical Society, India. Conference abstracts, Conference of the Free Radical Society, 2007. , Jan-2007
Abstracts for Journals and Proceedings	Keszenman D, Sutherland BM. "Analysis of DNA damage spectra induced by irradiations with the same HZE ion and different energies." International Congress of Radiation Research, San Francisco CA, July 8-12, 2007. Abstracts, International Congress of Radiation Research, San Francisco CA, July 2007, p. 216. , Jul-2007

Abstracts for Journals and Proceedings	Keszenman DJ, Sutherland BM. "Heavy ion-induced DNA damage: LET, Z or energy--that is the question." 6th International Conference of Biological Physics, Montevideo, Uruguay, August 2007. Conference abstracts, 6th International Conference of Biological Physics, Montevideo, Uruguay, August 2007. , Aug-2007
Abstracts for Journals and Proceedings	Sanchez AG, Keszenman DJ, Sutherland BM. "Induction of clustered DNA damage by gama-rays in Saccharomyces cerevisiae." 6th International Conference of Biological Physics, Montevideo, Uruguay, August 2007. Conference abstracts, 6th International Conference of Biological Physics, Montevideo, Uruguay, August 2007. , Aug-2007
Abstracts for Journals and Proceedings	Keszenman DJ, Sanchez AG, Sutherland BM. "Repair kinetics of DNA strand breaks and oxypurine clusters induced by gamma-radiation oxidative stress in Saccharomyces cerevisiae." Free Radicals in Montevideo 2007. V Meeting of the Society for Free Radicals, Biology and Medicine, Montevideo, Uruguay, September 2007. Conference abstracts, Free Radicals in Montevideo 2007. V Meeting of the Society for Free Radicals, Biology and Medicine, Montevideo, Uruguay, September 2007. , Sep-2007
Articles in Peer-reviewed Journals	Zhou G, Bennett PV, Cutter NC, Sutherland BM. "Proton-HZE-particle sequential dual-beam exposures increase anchorage-independent growth frequencies in primary human fibroblasts." Radiat Res. 2006 Sep;166(3):488-94. PMID: 16953667 , Sep-2006
Articles in Peer-reviewed Journals	Bennett PV, Cutter NC, Sutherland BM. "Split-dose exposures versus dual ion exposure in human cell neoplastic transformation." Radiat Environ Biophys. 2007 Jun;46(2):119-23. PMID: 17256176 , Jun-2007
Articles in Peer-reviewed Journals	Paap B, Wilson DM 3rd, Sutherland BM. "Human abasic endonuclease action on multiple-lesion abasic clusters: implications for radiation-induced biological damage." Nucleic Acids Research, 2008 Mar 19; [Epub ahead of print] PMID: 18353858 , Mar-2008
Books/Book Chapters	Sutherland BM, Bennett PV, Sutherland JC. "DNA damage quantitation by alkaline gel electrophoresis." in "Molecular Biology Methods: DNA Repair, 2nd Ed." Ed. D. Henderson. Methods Mol Biol. 2006;314:251-73. PMID: 16673887, Dec-2006